

AR Mount Guide

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01. Introduction

The AR Mount Guide is an innovative mobile application designed to revolutionize how people explore and understand mountain landscapes. Using the power of augmented reality (AR), the system enables users to identify nearby peaks instantly by simply pointing their smartphone toward the horizon. The app overlays real-time mountain labels and detailed information directly onto the camera view, creating an immersive, intuitive, and educational outdoor experience. In addition, when clicked on the detected mountain, the application displays the 3d model of mountain as well as detailed description.

02. Aims & Objectives

The primary aim of this study is to research how modern technology like augmented reality (AR) can enhance mountain exploration for tourists and for regular people. It aims to increase the overall experience of users by creating an easy-to-use, cost-efficient, and more user-friendly AR mobile application.

The main objectives of this study are as follows:

- To research and implement a fully functional AR mobile application for overlaying the mountain and its information.
- To enhance an immersive and enjoyable experience by providing indoor AR simulation.
- To provide an optimized 3D model display of mountains with its detailed information.

03. Problem Statement

Many people still rely on tour guides, books, or online searches to learn about mountains, but these methods can be slow, unclear, and inconvenient. As a nature lover, I often struggled to identify the mountains I could see from my balcony, which inspired me to create a simple, accurate solution. My interest in augmented reality—sparked by apps like Instagram and Snapchat—motivated me to build an AR-based tool using Flutter, a framework I am already familiar with. I chose this project to expand my knowledge of AR, geolocation, and device orientation while solving a real-world problem. In the end, this app aims to make learning about mountains faster, easier, and more engaging.

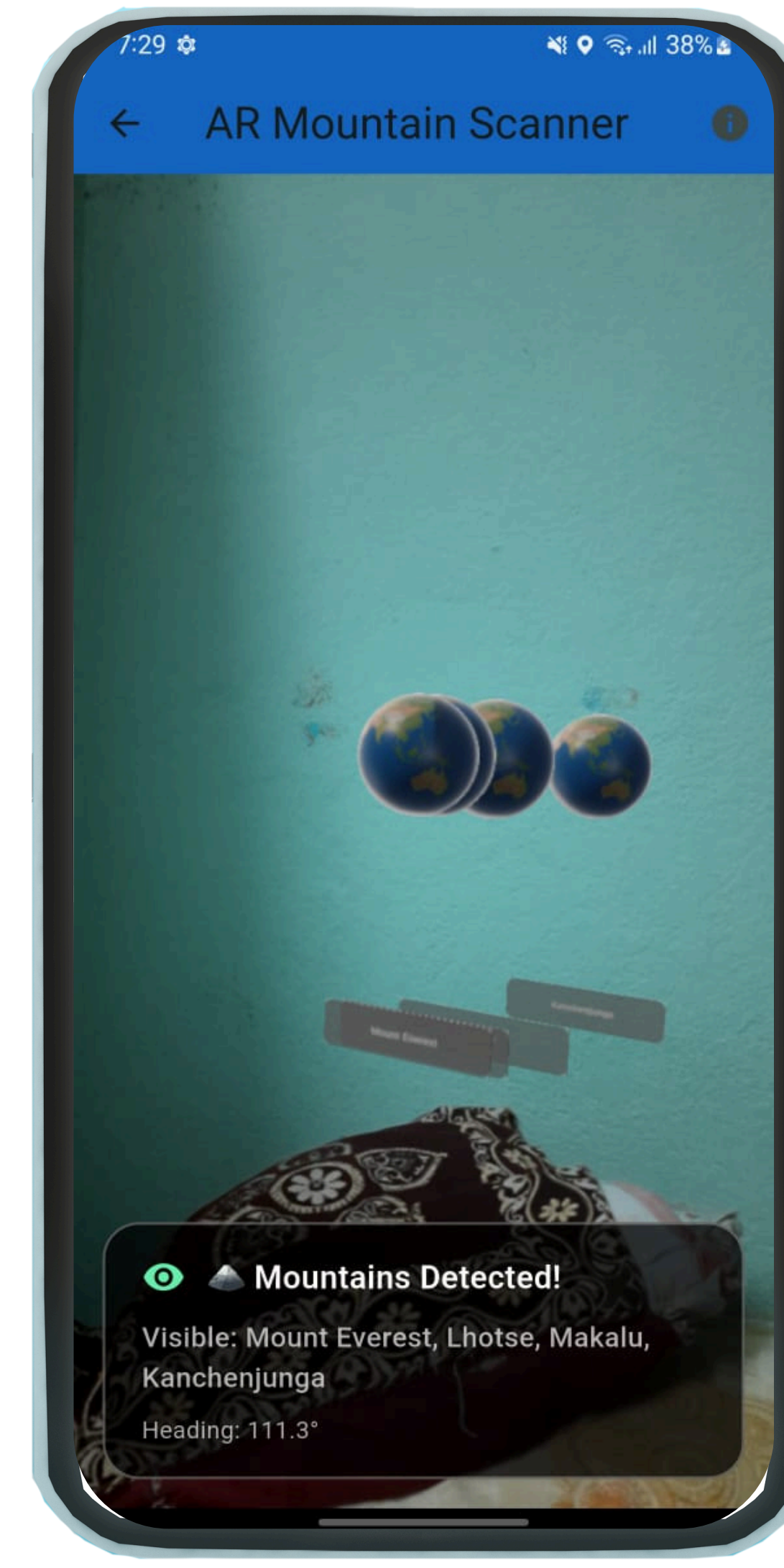
06. Key Features

- Authentication: The system allows users to create a new account and also log into an existing one
- Normal Overlay Screen: Users can scan mountains in normal mode where detected mountain will be displayed in text labels
- Ar Overlay Screen: Users can scan mountains, and when a mountain is detected, a 3D AR model along with its label will appear on the screen
- Mountain Detailed Screen: When tapped on detected mountain, user can view the 3d model of mountain along with its necessary information

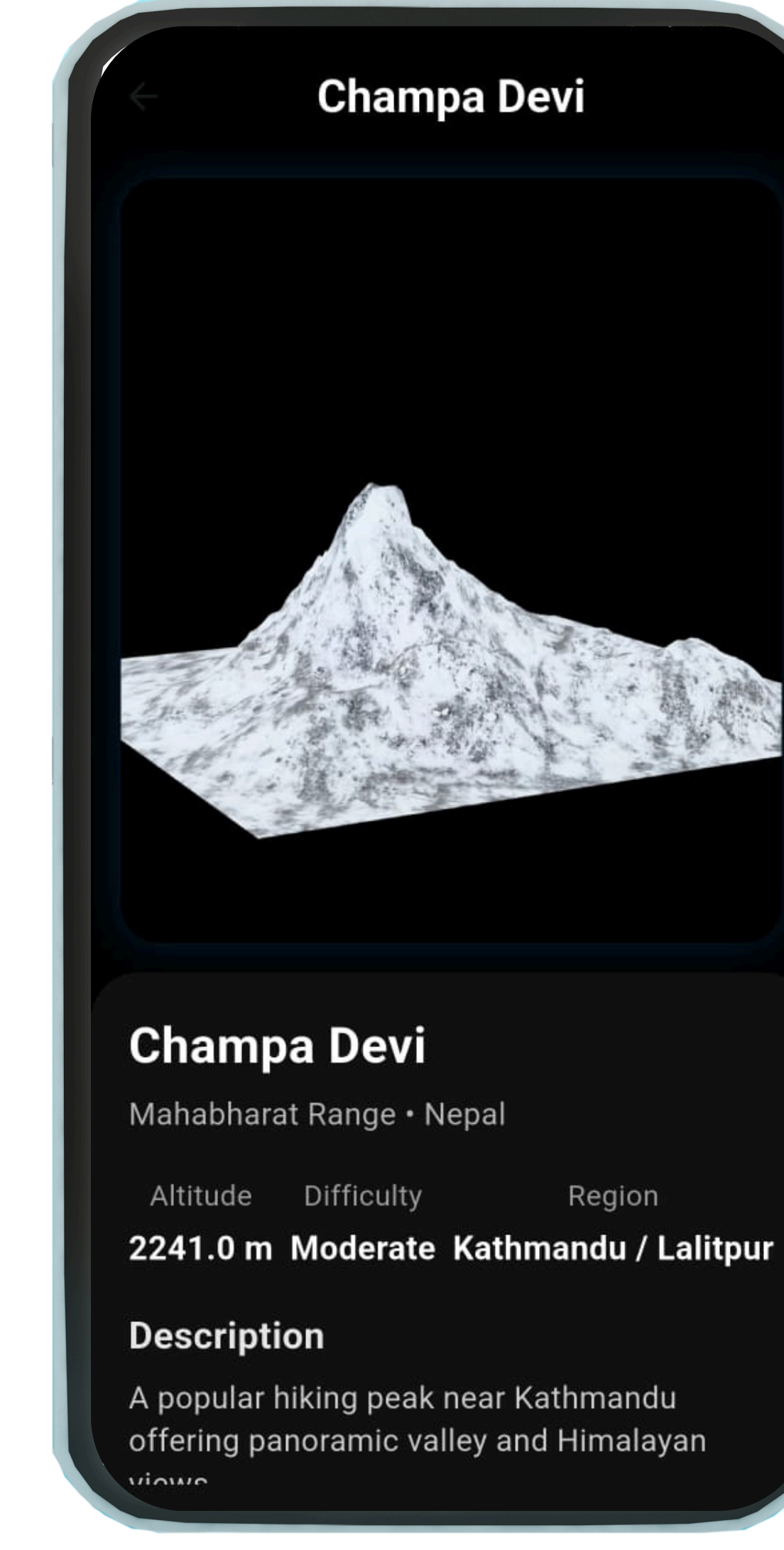
04. Graphical User Interface



Normal Simulation



Ar Mountain Screen



Mountain Detail Screen

05. Implementation

Methodology

The given project makes use of the fundamental idea of the agile approach, which facilitated the iterative development process. Augmented reality, sensor integration, real-time computations, and AR overlay are all part of the project. Agile approach greatly aids in dealing with it by enabling prompt adaption to any technological obstacles and by providing ongoing supervisory input.

Software Used

- Flutter for mobile application
- Node.js and express for backend server
- Mongo db as database

Hardware Used

- Smart phone with google ar core services installed

Key Packages Used

- camera: ^0.11.0+2
- sensors_plus: ^6.1.1
- geocoding: ^4.0.0
- logger: ^2.6.2
- geolocator: ^13.0.2
- flutter_compass: ^0.8.1
- model_viewer_plus: ^1.9.3
- http: ^1.4.0
- arcore_flutter_plugin:
- permission_handler: ^12.0.1
- vector_math: ^2.1.4

07. Reflective Analysis

Working on this project has significantly broadened my understanding of modern technologies, especially augmented reality, and how they can be applied to real-world challenges. Throughout development, I explored how AR overlays can identify mountains, strengthening my ability to connect emerging technology with practical use cases. I also gained hands-on experience integrating AR with geolocation, compass data, and device sensors, helping me understand how these systems work together to create accurate and interactive user experiences. Beyond the technical aspects, this project enhanced my problem-solving abilities, as I often had to troubleshoot issues related to sensor accuracy, AR rendering, and data alignment. It also improved my research skills, as I studied various AR frameworks, optimization techniques, and navigation concepts to implement effective solutions. Overall, this project deepened my technical knowledge and strengthened my analytical thinking, research capability, and confidence in building innovative, user-focused applications

08. Conclusion

In conclusion, the AR Mount Guide project successfully demonstrates how augmented reality and sensor-based technologies can enhance real-world exploration by making mountain identification more intuitive and engaging. Through the integration of AR overlays, geolocation, and sensor data, the system provides users with a seamless and informative experience that bridges technology with nature. This project not only achieves its intended goal but also showcases the potential of modern digital tools to solve practical problems, inspire learning, and elevate the way people interact with their surroundings. Moreover, it highlights how immersive technologies can transform outdoor experiences by offering users immediate, context-aware insights. Ultimately, the project serves as a foundation for future innovations in AR-based navigation, environmental education, and interactive exploration.

09. References

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